

ChokePilot — Autonomous Choke Controller

Honeywell Campus Connect Hackathon (PS3)

Speaker notes are in *italics* under each slide. Numbers below are pulled from one fresh run of `identify.py` → `verify_identification.py` → `scenarios.py` → `baselines.py` → `seed_sweep.py` → `scenario_d.py`, cross-checked against `docs/report.md` — “§N” references point back there for the full derivation.

Section 1 — Process Understanding & Model

Slide 1 — The Problem

Operators and engineers don’t under-perform because the physics is hard — they under-perform because of five well-documented, unglamorous frictions:

1. **Operators “baby” the choke** — fear of sand/formation damage leaves real production capacity unused. (*Eng-Tips forum discussion*)
2. **Too many wells, too little attention** — a choke gets set once during a shift and left alone, because nobody has the bandwidth to revisit it continuously. (*patent literature on standard industry practice*)
3. **Slugging is caught reactively** — a human notices fluctuations after the fact, not before. (*SPE technical paper*)
4. **APC doesn’t scale upstream because models need a dedicated team** — the on-record barrier isn’t algorithmic sophistication, it’s who keeps the model current. (*on-record Honeywell upstream solutions interview*)
5. **Alarm fatigue and distrust of opaque automation** — operators override what they don’t understand. (*NTSB SCADA safety review*)

ChokePilot answers all five: continuous optimization instead of a “set and forget” choke; a search-based, no-heavy-optimizer controller with a fresh from-scratch identification step instead of a model a dedicated team must maintain; and a logged one-sentence rationale for every move instead of a black box. It’s a small-scale, single-well version of the same closed-loop direction Honeywell’s APC customers are already moving toward on wells — a focused proof-of-concept of the wellhead-to-control-room optimization problem Honeywell’s Upstream Production Performance Suite (UPPS) addresses at scale. It does not claim equivalence; it echoes the same idea at a scale one person can build and verify in a project timeline. (report.md §0)

Slide 2 — No Official Simulator: What We Built, and What It Can't Tell Us (*report.md §1.1*)

- The platform provided no simulator — only `Autonomous_Choke_Control_Simulated_Dataset.csv` (120 hr, choke tested 30–65%) and a presentation template.
- `data/simulator.py` is a **calibrated substitute**, clearly commented as such, exposing the exact interface the brief specifies: `Q, WHP, FLP, BHP = simulator.step(choke_position)`.
- Steady-state map per channel: $y_{ss}(u) = A + B \cdot u / (u + u_h)$; Q is forced through zero at 0% choke (a fully shut choke physically gives zero flow).
- Dynamics: FOPDT, discretized with the **exact** zero-order-hold form `alpha = 1 - exp(-Ts/tau)` — not the Euler approximation `Ts/tau`, which is only accurate when `Ts << tau` and can go unstable outside that regime.
- The steady-state fit uses **only the last ~6 hours of each dwell period** (near-settled samples), not the whole trajectory — fitting on transient-contaminated samples was found to bias the curve and the downstream τ/θ search that compensates for it.

Two honestly-reported findings, not hidden: - With the `uh` search grid widened to check for boundary-pinning, **WHP, FLP, and BHP all pin at the widened boundary** — their steady-state response is statistically indistinguishable from linear over the tested 30–65% range. Only Q, anchored through zero, genuinely needs the saturating shape. - The reference CSV covers **only 5 distinct choke levels**, fit as a single deterministic least-squares point estimate — no confidence interval or uncertainty quantification anywhere in this pipeline.

Slide 3 — Open-Loop Step Test: What It Actually Tests (*report.md §1.2*)

- Per the brief, `identify.py` runs its **own** fresh monotonic staircase (0→100% in 10% steps, 40h dwell) against the calibrated simulator, rather than reusing the reference CSV.
- **Reframing an earlier claim:** `identify.py` imports `_fit_fopdt` and `_simulate_fopdt` directly from `data/simulator.py` — the *same* private fitting functions used to calibrate the simulator itself. The functional form is shared **by construction**. Calling this “treating the simulator as a black box” overstates what’s actually being tested.
- What it legitimately tests: **parameter identification accuracy under a correctly-specified model**. `verify_identification.py` compares identified vs. true τ across 5 seeds (0, 1, 2, 7, 99):

Channel	True τ (h)	Mean identified τ (h)	Mean error	Error range
Q	5.00	4.95	3.0%	0.0–5.0%

Channel	True τ (h)	Mean identified τ (h)	Mean error	Error range
WHP	7.50	7.20	4.0%	0.0–6.7%
FLP	7.00	6.95	3.6%	0.0–7.1%
BHP	9.00	8.25	8.3%	2.8–13.9%

*Dead-time identification now matches the true θ **exactly in all 20/20 seed×channel combinations** — no residual off-by-one anywhere on this table. BHP is the largest remaining τ residual (8.3%); Slide 4 explains why this table looks different from an earlier draft.*

Slide 4 — Two Fixes That Explain the Identification Error (*report.md §1.2*)

Two separate, sequential, tested-not-hypothesized fixes sit behind Slide 3's table — worth naming both, since stopping after the first would have missed the second.

Fix 1 — dwell time (DWELL_HOURS 24→40). With a 24h dwell, mean τ error was Q 20% / WHP 16% / FLP 27% / BHP 31%, identified τ lower than true τ in **all 20/20** seed×channel combinations — a one-directional bias consistent with insufficient settling time. Raising the dwell to 40h confirmed that diagnosis for **three of four channels** (WHP 16.0%→4.0%, FLP 27.1%→13.6%, BHP 31.1%→7.8%) and *ruled it out* for Q (20.0%→19.0%, unchanged across every dwell length tested, 24–80h) — a genuine open question at that point, not folded into an explanation that didn't cover it.

Fix 2 — a one-sample lag between the fitting code and the live simulator. `_simulate_fopdt` drove `sim[k+1]` from `y_ss[k]` (i.e. `u[k- $\theta$ ]`), but `Simulator.step()` actually drives the arriving sample from `u[t- $\theta$ ]` — the same time index, not one behind. A real structural bug, not a hypothesis, confirmed by tracing both code paths by hand. Fixing it (`y_ss[k] → y_ss[k+1]`) is what **resolved Q's open question**: error dropped 19.0%→3.0%, and θ went from matching exactly for only 2/4 channels to matching exactly for **all 4**. It also revealed the simulator's own "ground-truth" θ values had carried the identical bias, so recalibrating shifted the *true* θ by +1 across the board (τ unchanged) — both sides of every earlier comparison had been shifted the same way.

Net effect: BHP is now the largest residual (8.3%), not Q — the direct consequence of Fix 2 landing squarely on Q's bias.

Slide 5 — Hybrid Physics + Learned Correction: Three of Four Channels Earn Their Place (*report.md §1.3*)

- A small degree-1 polynomial-in-choke correction on the physics FOPDT residual, validated on a **held-out** step test before being trusted.

Channel	Physics-only RMSE	+Correction RMSE	Kept?
BHP	9.85	9.81	✔ used
FLP	0.92	0.89	✔ used
Q	1.66	1.63	✔ used
WHP	1.30	1.30	✗ skipped

This table changed shape after Slide 4’s Fix 2: with the old, laggy fit, only BHP’s correction generalized to held-out data. Post-fix, Q, FLP, and BHP all keep a small but consistent held-out improvement; only WHP’s correction makes things very slightly worse on held-out data (1.296→1.304) and is skipped.

`select_beneficial_corrections()` makes this call automatically per channel from held-out RMSE alone — nothing hand-picked, which is also why the verdict flipped for three channels without anyone hand-editing it. BHP remains both the largest τ residual (Slide 3) and the largest correction RMSE — plausibly absorbing some of that dynamics mismatch; this pipeline can’t distinguish the two.

Slide 6 — Safety Limits Are One-Sided, Not Symmetric Bands (report.md §1.4)

Channel	Direction enforced	Limit	Brief basis
WHP	floor only (<code>hi = +inf</code>)	≥ 205 psi	“If WHP becomes too low, the well may operate outside its recommended operating envelope.” High WHP just means the choke is closed back further — safe.
BHP	floor only (<code>hi = +inf</code>)	≥ 2830 psi	“one of the most important indicators of reservoir health and drawdown” — low BHP means excessive drawdown; high BHP means safely choked back.
FLP	ceiling only (<code>lo = -inf</code>)	≤ 200 psi	“helps ensure stable transportation of produced fluids” — the risk is backpressure/separator

Channel	Direction enforced	Limit	Brief basis
			overpressure on the high side, not a low reading.

An earlier draft bracketed all three symmetrically, which could reject a high-WHP/high-BHP or low-FLP state that isn't actually unsafe. These are further tightened inside the controller (Slide 8) before use.

Slide 7 — Monitored, Not Constrained: WHT & AP (*report.md §1.5*)

- `data/simulator.py` also simulates **Wellhead Temperature (WHT)** and **Annulus Pressure (AP)** — plotted (greyed out) in every scenario figure, but they **never feed the controller**.
- They exist because the brief lists them as part of “a complete production operating envelope” an operator would want visibility into — not because this challenge’s control problem is defined over them.
- WHT declines monotonically with choke opening (Joule-Thomson cooling); AP is flat, decoupled from choke position, carrying only an illustrative integrity-alarm band (1650–1950 psi) for situational awareness.
- Neither has a reference-CSV column to calibrate against, so unlike Q/WHP/FLP/BHP their curve parameters are hand-set placeholders — chosen to be qualitatively right, not fit to data.

Deliberate design line: surfaced for situational awareness — echoing Honeywell Asset Sentinel's risk/alert layer — but a WHT/AP reading can never change which choke move the controller picks.

Section 2 — Control Strategy

Slide 8 — What This Actually Is: One-Step Receding-Horizon Search, Not Full MPC (*report.md §2.1, §2.2*)

An earlier draft called this “brute-force MPC.” That overstates it.

Each control interval:

1. Enumerate **11 legal choke moves** (–5% to +5% in 1% steps) for **this interval only** — not a search over a sequence of future moves.
2. For each candidate, **hold that new position constant** for the rest of the lookahead horizon and simulate forward — a cheap proxy for “what happens if I stop adjusting,” not a genuine multi-step trajectory optimization.

3. Reject any candidate predicted to breach a WHP/FLP/BHP limit anywhere in that held-constant horizon. Limits are backed off by 3σ of each channel's identified `noise_std` first (standard robust-MPC constraint tightening) — noise-free predictions would otherwise ride the true limit and let real sensor noise breach it.
4. Command the feasible candidate whose end-of-horizon predicted oil rate is closest to target, ranked with a move-suppression penalty (Slide 14) so near-ties don't get decided by noise; if none is feasible, fall back to the one minimizing predicted violation, same penalty applied — the controller never freezes or refuses to act.
- This **is** receding-horizon in the classic sense — it re-runs every interval with fresh measurements — but the per-step search itself is a one-step decision screened by a hold-constant forward simulation. Horizon is derived, not hand-tuned: `ceil(3 × max( $\tau$ ) /  $T_s$ )`, clipped to [3, 12] h.
 - **Safety here is best-effort, not formally guaranteed** — no recursive feasibility check, no terminal invariant set. The 30-seed sweep (Slide 15) shows the safety-fallback branch firing on **23.6%** of Scenario C's steps — precisely because that scenario runs at the edge of what the model considers feasible, with no formal guarantee behind it, only a greedy per-step search that has worked well empirically.

Slide 9 — Scenario A's Violations: A Deterministic Initial-Condition Property, Not an Extrapolation Artifact (*report.md §2.3*)

Two earlier drafts got this wrong: the first claimed 15% choke was “inside supported territory” (false — the CSV's tested band is 30–65%); the second correctly retracted that but blamed the violations on generic “extrapolation uncertainty,” implying the numbers might not be real. Checking the model's own numbers directly says otherwise.

Computing the identified FLP steady-state map at low choke (the exact function the controller predicts with):

Choke	FLP steady-state	vs. 200 psi ceiling
0%	218.4 psi	exceeds
15%	203.7 psi	exceeds
20%	198.8 psi	clears (barely)
25%+	193.9 psi	clears with margin

Scenario A starts at 15% choke, and `Simulator.reset()` initializes FLP at exactly that choke's steady state — 203.7 psi, already over the ceiling, before the controller has made a single decision. With the $\pm 5\%$ /interval ramp limit, reaching even the steady-state-safe 20% choke takes one full interval,

and reaching the controller's own tightened-margin-safe $\sim 22\%$ takes two — a hard constraint the controller has no authority to override, plus FLP's own $\tau=7h/\theta=1h$ lag on top.

This is the identified model's own curve, trusted exactly as identified — not an artifact of guessed, untrustworthy numbers outside the calibrated range. A better 0–30% calibration could shift the exact numbers; it would not change the qualitative outcome unless it also flipped the sign of FLP's slope near 15%, which nothing in the physics suggests.

Slide 10 — Explainability Is a Logged Field, Not a Slide (*report.md §2.4*)

Directly answers pain point #5: every decision produces a real, human-readable rationale — a field in the output CSV, in both branches.

"Moved choke to 34.0% because it keeps WHP/FLP/BHP within safe limits over the next 12h and brings predicted oil rate to 99.4 bbl/hr, closest to the 100.0 bbl/hr target among 11 feasible options." — normal branch, actual logged decision

"No choke move keeps all limits satisfied over the lookahead; moved to 62.2% because it minimizes the predicted constraint violation (6.8 psi-steps over horizon)." — safety-fallback branch, same field, same guarantee

Commanded choke values are rounded to 0.1% before being used for prediction, selection, and the logged string — the rationale can never silently diverge from what was actually applied.

Honeywell echo: dashboard/output layering mirrors Uniformance's PHD (historian) / KPI (target-tracking) / Asset Sentinel (risk/alert) structure — a focused proof-of-concept of the same wellhead-to-control-room optimization problem UPPS addresses at scale. (*echoes, not replicates.*)

Section 3 — Results

Single-run numbers below use the default seed; the 30-seed sweep (Slide 15) is the number to trust over any one run.

Slide 11 — Scenario A: Startup to Target (*report.md §3.1*)

- 15% choke → 100 bbl/hr target, 80-hour run.
- **Tracking:** trailing 10-hour mean (hours 70–79) of **99.26 bbl/hr** at a steady 34.0% choke.
- **Safety, single seed:** 4/80. **30-seed sweep:** every seed shows at least one violation (**30/30**), mean **3.97/80**, max **5/80**.

- **Time to enter the safe envelope** (new metric, Slide 9's mechanism made measurable): single-seed **4h**; 30-seed sweep **mean 4.0h, range 3–5h** — tight enough to support “deterministic,” not “noisy extrapolation.”
- **Ramp-rate:** 0/80 violations in every seed — the ramp limit is the *reason* the envelope takes several hours to reach, not itself violated.

Slide 12 — Scenario B: Target Tracking (*report.md §3.2*)

- 34.2% choke start, target steps **100 → 150 bbl/hr at t = 60h**, 140-hour run.
- **Tracking:** trailing 10-hour mean **100.32 bbl/hr** at 34.2% choke before the step (hours 50–59); **150.57 bbl/hr** at 61.2% choke after settling (hours 130–139).
- **Safety:** 0/140 in the representative run **and 0/140 in all 30 seeds** — the only scenario with a perfectly clean sweep, because it never leaves the calibration's supported 30–65% band.

Slide 13 — Scenario C: Infeasible Target (*report.md §3.3*)

- 34.2% choke start, **400 bbl/hr requested** — deliberately beyond what's safely achievable.
- **Behavior:** does not chase the target into a violation. Trailing 10-hour mean **162.76 bbl/hr** at 69.4% choke — the maximum rate the tightened envelope allows, not 400.
- **Safety:** 0/100 single-seed, **and 0/100 in all 30 seeds** — fully clean.
- **Safety-fallback frequency: 23.6%** of steps — far higher than A or B, because this scenario deliberately runs the choke near the edge of the tightened feasible envelope for its entire duration. Frequent fallback here is expected behavior, not a red flag: 0 actual violations means the fallback branch is doing its job.

Slide 14 — Actuator Activity: Fixing the Chattering (*report.md §3.4*)

The controller originally picked the feasible candidate purely by predicted target error, with move size only a last-resort tie-break. Near target, many candidates' errors differ by less than measurement noise — the “closest” one was effectively a coin flip, so the valve chattered chasing error it can't actually predict.

Fix: move-suppression cost on both branches — $\text{cost} = q_err + \lambda \cdot |\Delta u|$ (feasible) / $\text{cost} = \text{violation} + \lambda \cdot |\Delta u|$ (fallback), $\lambda = 1.0$ bbl/hr of predicted improvement required per %-point moved.

Scenario	Moves	Total valve travel
A	5 / 80	16.0 %-pts
B	7 / 140	29.0 %-pts

Scenario	Moves	Total valve travel
C	53 / 100	129.0 %-pts

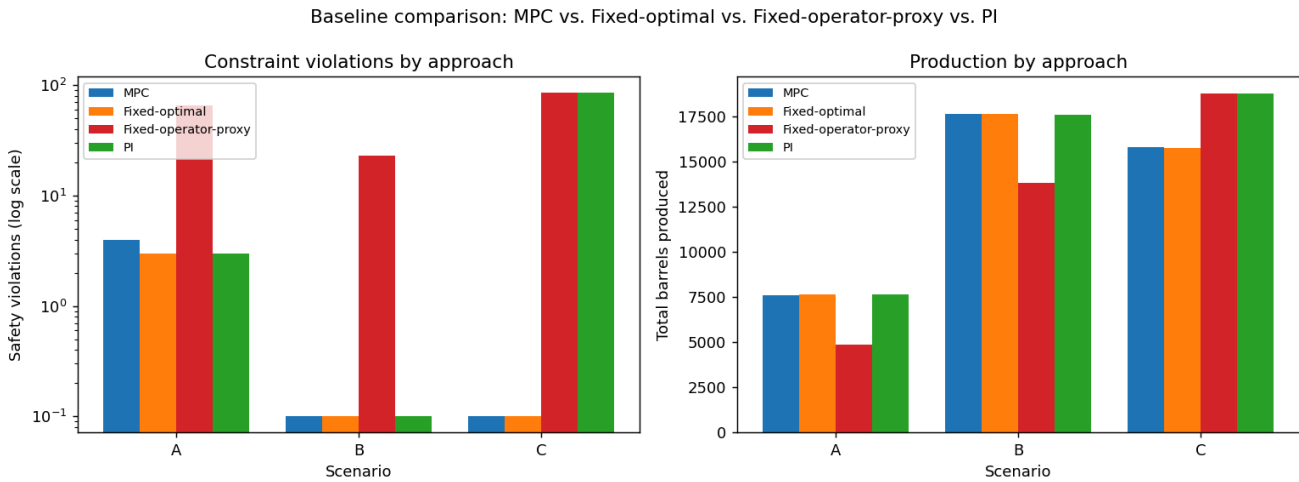
Honest result, not fully positive: the fix works as intended for A and B — 5 and 7 moves total, vs. constant hunting beforehand. **Scenario C is only partially improved:** travel dropped 16% (154→129 %-pts) but move count is essentially unchanged (52→53). Root cause: C rides the tightened WHP floor with real measurement noise ($\sigma \approx 1.2$ psi) that flips **which candidates are feasible at all**, step to step — a ranking fix within the feasible set can't fix the feasible set itself changing.

Slide 15 — Seed-Sweep Distribution: 30 Seeds per Scenario (report.md §3.5)

Scenario	Mean violations	Max violations	Seeds with ≥ 1 violation	Mean safety-fallback rate
A — Startup to Target	3.97 / 80	5	30 / 30	8.58%
B — Target Tracking	0.00 / 140	0	0 / 30	0.00%
C — Infeasible Target	0.00 / 100	0	0 / 30	23.57%

Full per-seed results: `outputs/seed_sweep_results.csv`. A single seed's "0 violations" is a point sample from a noisy system — this distribution, not any one run, is what actually supports a safety claim.

Slide 16 — Baseline Comparison: MPC vs. Fixed-Optimal vs. Fixed-Operator-Proxy vs. PI (report.md §3.6)



Baseline comparison: constraint violations (log scale) and total barrels produced, per approach, per scenario

- **Fixed-optimal** — the choke the identified model says holds the target at steady state, walked back until its own steady-state predictions clear the tightened envelope, then held. Model-informed, envelope-aware — what a good engineer *with* this pipeline would set.
- **Fixed-operator-proxy** (new) — **no model, no envelope knowledge**: a naive straight-line read of choke-vs-oil-rate off the raw reference CSV (the only data a real operator without this pipeline would have), backed off 15 points. Models pain point #1 directly.
- **PI** — velocity-form PI on oil rate, IMC-tuned from the identified model, blind to WHP/FLP/BHP by design.

Scenario	Approach	Safety violations	Total barrels
A	MPC	4/80	7,590.4
A	Fixed-optimal	3/80	7,657.6
A	Fixed-operator-proxy	66/80	4,852.4
A	PI	3/80	7,652.1
B	MPC	0/140	17,676.4
B	Fixed-optimal	0/140	17,642.0
B	Fixed-operator-proxy	23/140	13,824.5
B	PI	0/140	17,591.0
C	MPC	0/100	15,806.2
C	Fixed-optimal	0/100	15,769.0
C	Fixed-operator-proxy	85/100	18,778.3
C	PI	85/100	18,778.3

*The honest headline — not the one Scenario D (Slide 17) was built to find: **MPC ties Fixed-optimal in every scenario** (no cross-channel coupling for live re-planning to exploit here). The real, consistent win is **envelope-awareness itself vs. the operator-proxy**, in all four scenarios — closing the choke 15% below naive belief looks conservative but actually pushes FLP the wrong* direction, past its ~19% threshold (Slide 9), which a model-blind operator has no way to know.**

Slide 17 — Scenario D: Disturbance Rejection (*report.md §3.6*)

Built specifically to find MPC's edge: does live re-planning beat a static setpoint once the plant genuinely drifts underneath both? A deliberate, explicit relaxation of the brief's "no changing reservoir properties" simplification.

- BHP's identified steady-state offset drifts **-0.5 psi/h for 200h** (reservoir decline) at a constant 100 bbl/hr target. The identified model is fit once, before the disturbance starts, and never updated.

Approach	Safety violations	Total barrels	Time-to-first-violation
MPC	0/200	20,040.9	never
Fixed-optimal	0/200	20,048.3	never
Fixed-operator-proxy	121/200	12,594.2	21h

Told straight, not spun: MPC and Fixed-optimal **tie** — 0 violations each. In this model, BHP and Q are independent, uncoupled FOPDT channels, so a declining reservoir never moves Q off target and never drifts BHP close enough to its floor (ends ~170 psi above it after 200h) to matter. MPC's re-planning had nothing to correct that a static, envelope-aware setpoint didn't already handle. Fixed-operator-proxy still fails here — but for the *same static-FLP-threshold reason as Scenario A/Slide 9*, not because it missed the disturbance: violations start at hour 21 (as soon as the ramp-limited approach reaches its 18.7% setpoint) and continue at a roughly steady rate for all 200h.

Slide 18 — Lessons Learned (*report.md §3.7*)

- **“Extrapolation” was a hedge, not the actual explanation.** Checking the model's own FLP steady-state curve (Slide 9) found a concrete, checkable fact — it exceeds 200 psi below ~19% choke — not a hedge about what an uncalibrated region *might* do.
- **A correctly-specified model still shows real identification error, and two separate root causes were both worth chasing down, not just naming one.** Dwell time (Slide 4, Fix 1) explained three of four channels and correctly *ruled itself out* for Q; a one-sample lag (Fix 2) was Q's actual answer, found by refusing to let Fix 1's partial success explain away the residual it didn't touch.
- **Name the controller accurately.** Calling this “brute-force MPC” implied guarantees — recursive feasibility, trajectory optimality — it doesn't have. It's a one-step receding-horizon search with hold-constant prediction: empirically strong on safety, but not formally proven.
- **Report distributions, not single runs.** A single seed's “0 violations” is a point sample. The 30-seed sweep is what actually supports a safety claim or a tracking claim.
- **A correction layer earns its place by generalizing, channel by channel.** Post-fix, it helps three of four channels (Q, FLP, BHP) and is skipped only for WHP — the verdict changed entirely from an earlier draft without anyone hand-editing which channels are marked used.
- **Chattering has more than one cause, and a single fix doesn't necessarily cure both.** Move-suppression cost fully fixed A/B (noise-tie chattering) but only partially fixed C, because C's chattering comes from the *feasible set itself* flipping under measurement noise — a different mechanism the same fix can't fully reach.
- **The real win is envelope-awareness, not live re-planning — say so even though it complicates the headline.** Scenario D was built to find MPC's edge over a static setpoint and didn't find one

(they tie). The consistent, honest win is against the operator-proxy baseline, in all four scenarios — a less dramatic but more accurate story than the one the experiment was designed to produce.